

FICTIONAL DEMONSTRATION DATA - NOT REAL FACILITY DATA
=====

1. EXECUTIVE SUMMARY

Verification score: 94.0%
Evidence-backed: 8/9
Model routing: Qwen3-4B (reasoning)

2. FINDINGS

1. [HIGH] Potential corrosion detected around P-101 outlet
Equipment: P-101 | Confidence: 92.0%
Ultrasonic and visual inspection indicate wall-thickness loss near the P-101 discharge outlet, approaching minimum allowable.
- Evidence: Inspection_Report_2026.pdf (p.7) [92.0%]
"Minor corrosion observed near the outlet section of feed pump P-101; wall thickness measured at 6.8 mm, minimum allowable is 6.4 mm.
- Evidence: Engineering_Calculation.pdf (p.3) [88.0%]
"With measured 6.8 mm and minimum allowable 6.4 mm, remaining life is approximately 1.3 years."
2. [MEDIUM] Control valve V-204 actuator sluggish response
Equipment: V-204 | Confidence: 86.0%
Valve V-204 exhibits delayed actuation; stem packing degradation is the likely cause per OEM guidance.
- Evidence: Inspection_Report_2026.pdf (p.8) [86.0%]
"Control valve V-204 actuator shows sluggish response; stem packing to be replaced at next turnaround.
- Evidence: Equipment_Manual.pdf (p.14) [83.0%]
"Control valve V-204 recommended packing replacement interval is 24 months."
3. [LOW] HX-301 fouling factor trending upward (needs human review) [NEEDS HUMAN REVIEW]
Equipment: HX-301 | Confidence: 82.0%
Heat exchanger HX-301 shows an upward fouling trend. Currently within limits but the trend is ambiguous.
- Evidence: Inspection_Report_2026.pdf (p.9) [82.0%]
"Heat exchanger HX-301 tube bundle fouling factor trending upward but acceptable."
4. [INFO] P-102 backup pump confirmed available
Equipment: P-102 | Confidence: 96.0%
Backup feed pump P-102 inspected and confirmed in standby-ready condition.
- Evidence: Equipment_Manual.pdf (p.5) [96.0%]
"Feed Pump P-101 and P-102 are horizontal centrifugal pumps rated 120 m3/h at 45 m head."

3. VERIFICATION

Score: 94.0% | Evidence-backed: 8/9
Findings flagged for review: HX-301 fouling factor trending upward (needs human review)

4. PIPELINE TRACE

- [security_agent] Security perimeter verified
- [router_agent] Task routed to Qwen3-4B
- [router_agent] Execution plan created

- [document_agent] Uploaded document loaded
- [document_agent] Document analysis prepared
- [vision_agent] P&ID regions detected
- [knowledge_agent] Knowledge retrieved
- [reasoning_agent] Findings generated
- [verification_agent] Evidence cross-check completed